Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Implicit functions and problems

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Implicit functions and problems

1 Implicit functions and problems

Study focus: Implicit functions and problems. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

It will be a continuation of chain rule and composite functions and other things that we saw in the previous video.

Key details

- which is like f of x y equal to c some constant, then we call this as an implicit function.
- This is an implicit function.
- I have x cube y plus 3xy square equal to 3.
- How do we differentiate this?
- You differentiate this using product rule, keep the first function as it is.
- Derivative of y is dy by dx, plus, keep the second function as it is, and you differentiate the first, which is 3x square, plus.
- You keep the first function as it is, you differentiate the second again, 2y dy by dx plus I'll keep the second function as it is and I differentiate.
- The advantage of this is the right hand side will become 0.
- Because it's a constant.
- And from this expression, it's easy for me to find what is dy by tx.
- or f of x y equal to some constant.
- But now if z is a function of xy, but with a restriction.
- That restriction you have to remember.
- That z must be a constant.
- So here if z is a function of n independent variable and if it's equal to constant, I can always find the total derivative of z which is defined by.
- DZ is equal to, so this, dz is equal to partial derivative with respect to x, dx1 with respect to just now we discussed what is the total derivative, right?

- If z becomes constant, then it looks like f of x comma y equal to c.
- This is called a level surface.
- And it's not new to you, right?
- Or isotherms where at every point the temperature remains the same.
- that this is nothing but a sphere centered at the origin, right?
- If C1 become 1, I'll get the unit sphere.
- C1 become 89, I'll get a bigger sphere.
- Every color would represent a different value of C, and it shows that for every color the temperature distribution or whatever you want to call it, the pressure or the potential or distribution of current, everything was remains constant.
- Of course, it's difficult to plot because I don't have that many dimensions to plot, but definitely it would represent a level surface.
- And in case of level surface, you can find the total derivative.

Example or application from the lecture

- For example, you can see here I have an problem x cube y plus 3xy square equal to 3.
- And there is a special name given to this.
- The level surface, for example, you take equipotential surfaces.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let me go back to my previous slide so that it will be easier for you.

Key details

- But is there another way that I can somehow make use of a partial derivative to this?
- If you observe carefully, you have this, right?
- Let me take this as some set.
- Isn't it ZX 3x square y plus 3y square?
- I am differentiating z partially with respect to x.
- And what is z y?
- Z y will be.
- Again I get x q plus.
- 6x y and if you observe carefully I got this dy by dx as minus zx by sec y.
- This will be useful if your function is too complicated.
- the see the tedious task that we did here I have to use product rule maybe quotient rule or chain rule or whatever it is the function might be as complicated as possible but it's not possible for me to find the d by dx very easily or very conveniently.
- But now if I need to find the total derivative, you already know z is a constant, so that's not going to change.
- That was a small observation.
- And that was a good observation.
- And this df is called the differential.

- In fact, you have already used this in your course.
- And where did we use this?
- How did you do this integration?
- As soon as you look at the integrand, it's clear that you have to make a substitution.
- That is where I can apply the total differential.
- That's mean basically I'm I'm in x plane, now I'll be in dt plane.
- All my x square x gets transformed and I'll get only a simple function as z to the power of t which you can integrate directly.
- Another appro application would be in errors and approximations that I'll discuss by the end of this unit when we do applications.
- These two are independent.
- those things you can find.

Example or application from the lecture

- I differentiate it given expression with respect to X and Y, and then this is the expression I got.
- Can you find out what is ZX from the given expression?
- And again, I need to consider that z is a function of xy, otherwise it's difficult for me to express it.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

Z y so this is like an extension of what we just discussed to another variable.

Key details

- One more variable is added.
- Again it represents a level surface.
- what I want you to concentrate is Zx.
- maybe I'll give you x cube y square plus three x y z plus some sine of e to the power of z this is equal to some c just imagine how much effort you need to put to differentiate this with respect to.
- X partially, right?
- You have to separate Z and then or because X and Y are both independent.
- You'll get you don't know how what to do with Y X and keep treat Y as a constant and retain that and everything.
- But here this formula will tell you that dou z by dou x is minus f x by f z.
- You can extend it to n number of variables.
- So the first question is for the curve x e power y plus y e power x equal to 0, find the equivalent of the tangent line.
- That means it is of the form z is equal to some function of xy or y x equal to zero.
- That's from the definition of total derivative.
- F of x comma y is x into e to the power of y plus y into e to the power of x.

- That's f x.
- F y will be x into e to the power of y plus e to the power of x.
- divided by X e to the power of Y plus e to the power of X.
- This term will vanish.
- The question says, find dy by dx if cos x par y is equal to sine y to the power of x.

Example or application from the lecture

- Suppose you have f of x y z equal to c, then zx is minus f x by f z.
- I know that the total derivative for this is given by so in this case if f of x y is equal to 0, then dy by dx is nothing but minus f x divided by f.
- If you need to find the tangent in any other point, let's say I want to find at the point x0 y naught, then you can write this as y minus y naught is equal to m into x minus x naught.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 15:00–19:59

Can we rewrite this?

Key details

- That means this represents an implicit function for me now.
- See, just imagine if you differentiate this directly, you need to differentiate like first cos x and then apply chain rule to this, it's to the power of y, and then sine y to the power of x.
- Things will become more complicated, right?
- But if we use partial derivatives, things will be much simpler.
- The formula for dy by dx will be minus fx divided by f y.
- When I differentiate this with respect to x, y is a constant.
- Is this clear what I did?
- So I had this as the first.
- First this is just the first term.
- Ah derivative of cos x is minus sin x.
- What is the derivative of a power x for you?
- If you remember derivative of a power x will be a to the power of x into ln of a.
- That's my numerator.
- That's the numerator, okay?
- So that means cos x will become a constant for me now.
- And derivative of a power y will be cos x to the power of y into ln of cos x.
- Minus same thing I have to do.
- This is a variable for me, but x is a constant.
- Of course, not much simplific simplification can be done for this, but this is the expression that you get for dy by dx.
- But you need to be careful.
- that x cannot take the value 0, neither y can take the value 0 because ln of 0 is not defined.
- So we'll try the next problem, which is again on implicit functions and we'll see how it goes.

- The question is find dou y by dou x and double at the point one comma zero comma two for this particular function.
- That's what is expected.
- And when I differentiate z partially with respect to x, I need to treat y as constant.
- derivative of y with respect to x by taking f minus of f x by f y.
- This is using the formula for total derivative.
- Where what is f?
- F is Z cube plus XY minus Y square Z.
- Please Z is constant.

Example or application from the lecture

- So let's differentiate the exponent first, which is y cos x to the power of y minus one.
- If any particular point is given, then we may continue and substitute that point and find out.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 20:00–24:59

I have two terms with y in it.

Key details

- Minus two Y Z.
- This is what I get.
- But I need to calculate y x at the point 1 comma 0 comma 2.
- At the point 1 comma 0 comet the value of y, y is 0 divided by x, x is 1 minus 2 into y is again 0 and z is 2.
- So that means I'll get this value as zero because the numerator is zero for me.
- Dose by dou x will be minus f x divided by f z.
- There's only one y term and f z will be three z square minus y square.
- I am differentiating f partial b with respect to z.
- But again I need to calculate this at the point.
- One comma, zero, comma two.
- So because y is zero, this is also zero for me.
- Tangent of the slope or the slope of the tangent at this point parallel to x axis is a horizontal line and even at this point it's a horizontal line.
- In our next session we will try some problems on Taylor's and McLaren series.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Implicit functions and problems transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00024	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice